Lesson 7: Probability Mass Functions (pmf's)

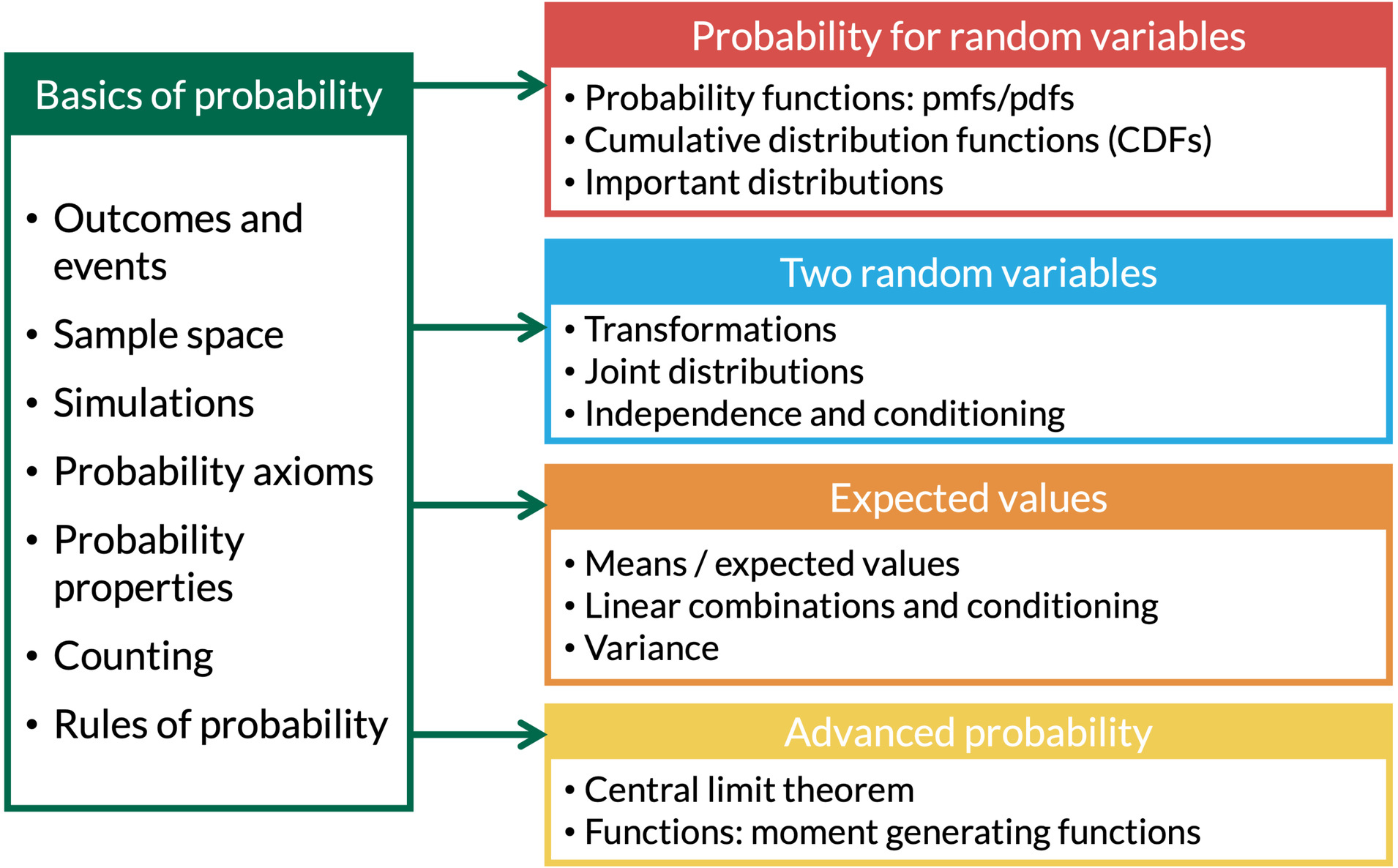
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Learning Objectives

1. Identify a probability mass function (pmf) from past simulations
2. Identify a binomial random variable and its parameters from a word problem
3. Use R to calculate probabilities and simulate binomial random variables

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

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From Lesson 2: Types of random variables

There are two types of random variables:

- **Discrete random variables** (RVs): the set of possible values is either finite or can be put into a countably infinite list
  - You could *theoretically* list the specific possible outcomes that the variable can take
  - If you sum the rolls of three dice, you must get a whole number. For example, you can't get any number between 3 and 4.
  - Described by a **probability mass function (pmf)**

- **Continuous random variables** (RVs): take on values from continuous *intervals*, or unions of continuous intervals
  - Variable takes on a range of values, but there are infinitely possible values within the range
  - If you keep track of the time you sleep, you can sleep for 8 hours or 7.9 hours or 7.99 hours or 7.999 hours …
  - Described by a probability density function (pdf)

What is a probability mass function?

Definition: probability distribution or probability mass function (pmf)

The **probability distribution** or **probability mass function** (**pmf**) of a discrete r.v. X is defined for every number x by

$$p_X(x) = \mathbb{P}(X = x) = \mathbb{P}(\text{all } \omega \in S : X(\omega) = x)$$

- X (capital) is the random variable, and x (lowercase) is one specific value that X can take
- Quick example: flip a fair coin once. Let $X = 1$ if the coin lands heads and $X = 0$ if it lands tails.
  - The pmf at $x = 1$ is $p_X(1) = \mathbb{P}(X = 1) = 0.5$

From Lesson 2: Simulating two rolls (1/2)

Example: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. Let X be the sum of the two rolls. How would we simulate X?

```
1  set.seed(4764)                                                    ①
2  reps <- 10000                                                     ②
3
4  rolls <- expand_grid(sim = 1:reps, die = 1:2) |>                  ③
5    mutate(roll = sample(1:4, n(), replace = TRUE))                 ④
6
7  X_sims <- rolls |>                                                ⑤
8    group_by(sim) |>
9    summarise(X = sum(roll))
```

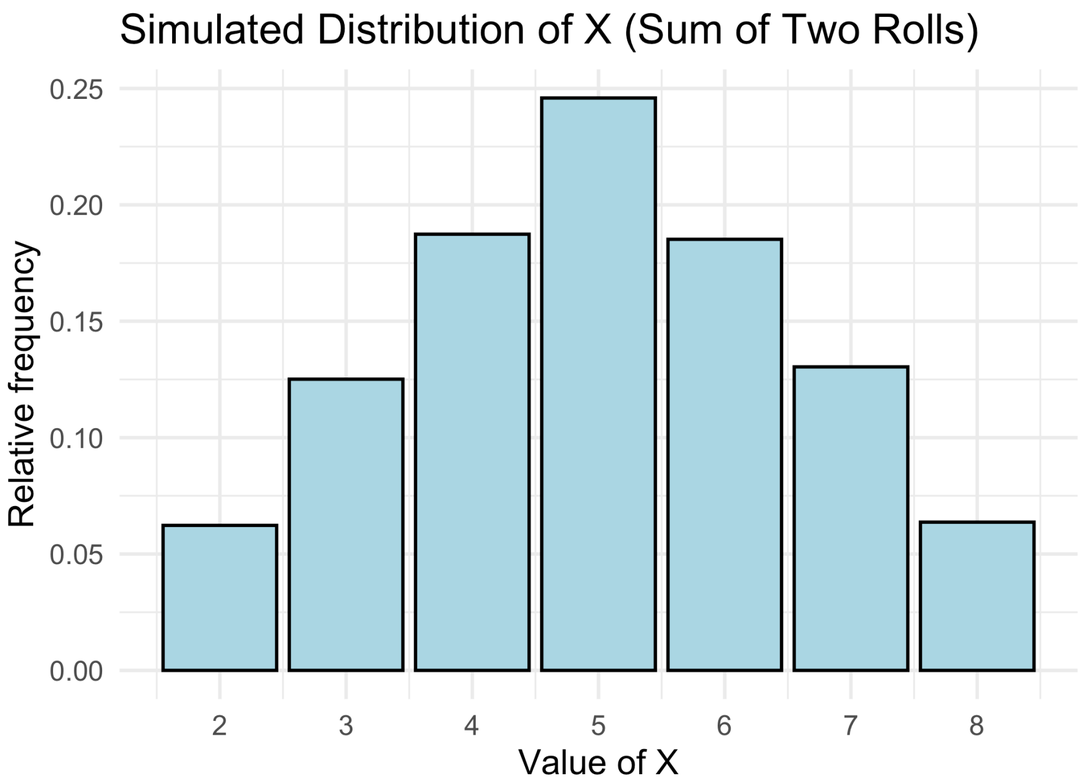
- The first few simulated values of X:

```
1  X_sims                                                            ①
```

```
# A tibble: 10,000 × 2
     sim     X
   <int> <int>
 1     1     4
 2     2     2
 3     3     5
 4     4     5
 5     5     4
 6     6     7
 7     7     5
 8     8     7
 9     9     6
10    10     2
# ℹ 9,990 more rows
```

From Lesson 2: Simulating two rolls (2/2)

- Plot simulated distribution of X



- Simulations give us the **approximate pmf** of X

- For the RV X, we can find the probability for each possible value, $P(X = x) = p_X(x)$:

$$p_X(x) = \begin{cases} \frac{4-|x-5|}{16}, & x = 2, 3, 4, 5, 6, 7, 8, \\ 0, & \text{otherwise} \end{cases}$$

- For any value that X cannot take, like $x = 2.5$, the pmf is 0

Remarks on the pmf

Properties of pmf

A **pmf** $p_X(x)$ must satisfy the following properties:

- $0 \leq p_X(x) \leq 1$ for all x
- $\sum_{\{all\ x\}} p_X(x) = 1$
  - Sum across all probabilities (for possible values) must be 1

- Some distributions follow the same patterns but change slightly depending on *parameters*
  - Each value of a parameter gives a **different** pmf
  - In previous example, the number of dice rolled was a parameter
    - We rolled 2 dice
    - If we rolled 4 dice, we'd get a **different** pmf!
  - The collection of all pmf's *for different values of the parameters* is called a **family** of pmf's

Learning Objectives

1. Identify a probability mass function (pmf) from past simulations

2. Identify a binomial random variable and its parameters from a word problem

3. Use R to calculate probabilities and simulate binomial random variables

Binomial random variables

- **One specific type of discrete random variable** is a **binomial random variable**

Binomial random variable

- X is a binomial random variable if it represents the number of successes in n independent replications (or trials) of an experiment where
  - Each replicate has two possible outcomes: either **success** or **failure**
  - The probability of success is p
  - The probability of failure is $q = 1 - p$

- A **binomial random variable** takes on values $0, 1, 2, \dots, n$.
- If a r.v. X is modeled by a Binomial distribution, then we write in shorthand $X \sim \text{Binom}(n, p)$
- *Quick example:* The number of heads in 3 tosses of a fair coin is a binomial random variable with parameters $n = 3$ and $p = 0.5$ (we count a head as a "success").

Binomial family of distributions

Distribution (or pmf) of a Binomial random variable

Let X be the total number of successes in n independent trials, each with probability p of a success. Then probability of observing exactly x successes in n independent trials is

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, 2, \dots, n$$

- The parameters of a binomial distribution are p and n.
- In the formula:
  - p is the probability of success for each trial
  - p^x is the probability of x successes, and $(1-p)^{n-x}$ is the probability of the $n-x$ failed trials
  - $\binom{n}{x}$ counts the ways to choose which x of the n trials are the successes

Binomial distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rbinom()`	returns sample of random variables with specified binomial distribution	simulations!
`dbinom()`	returns probability of getting certain number of successes	$P(X = x)$
`pbinom()`	returns cumulative probability of getting certain number or less successes	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qbinom()`	returns number of successes corresponding to desired quantile	Goes the other direction: we give it a probability p and it returns the value k where $P(X \leq k) = p$

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Falls in Older Adults (1/8)

Example 1: Falls in Older Adults

A major public health concern is falls among older adults (age 65+). National data suggests that 25% of older adults will experience at least one fall within a given year. A community health program is tracking a random group of $n = 8$ older adults for one year. Assume the likelihood of falling is independent from person to person.

Let X be the random variable representing the number of individuals in this group who experience at least one fall.

1. What is the sample space for the random variable X?
2. Write the probability mass function (pmf) for X.
3. Use R to calculate the probability for each possible value of X.
4. Make a bar plot of the pmf.
5. Simulate X for 10000 groups and plot the approximated pmf.

Falls in Older Adults (2/8)

Example 1: Falls in Older Adults

1. What is the sample space for the random variable X?

Falls in Older Adults (3/8)

Example 1: Falls in Older Adults

2. Write the probability mass function (pmf) for X.

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, 2, \dots, n$$

Falls in Older Adults (4/8)

Example 1: Falls in Older Adults

3. Use R to calculate the probability for each possible value of X.

```
1 n <- 8                                                          ①
2 p <- 0.25
3
4 dbinom(0, size = n, prob = p)                                   ②
5
6 falls <- tibble(                                                ③
7   x = 0:n,                                                      ④
8   prob = dbinom(x, size = n, prob = p)                          ⑤
9 )
```

```
[1] 0.1001129
```

```
1 falls                                                           ①
```

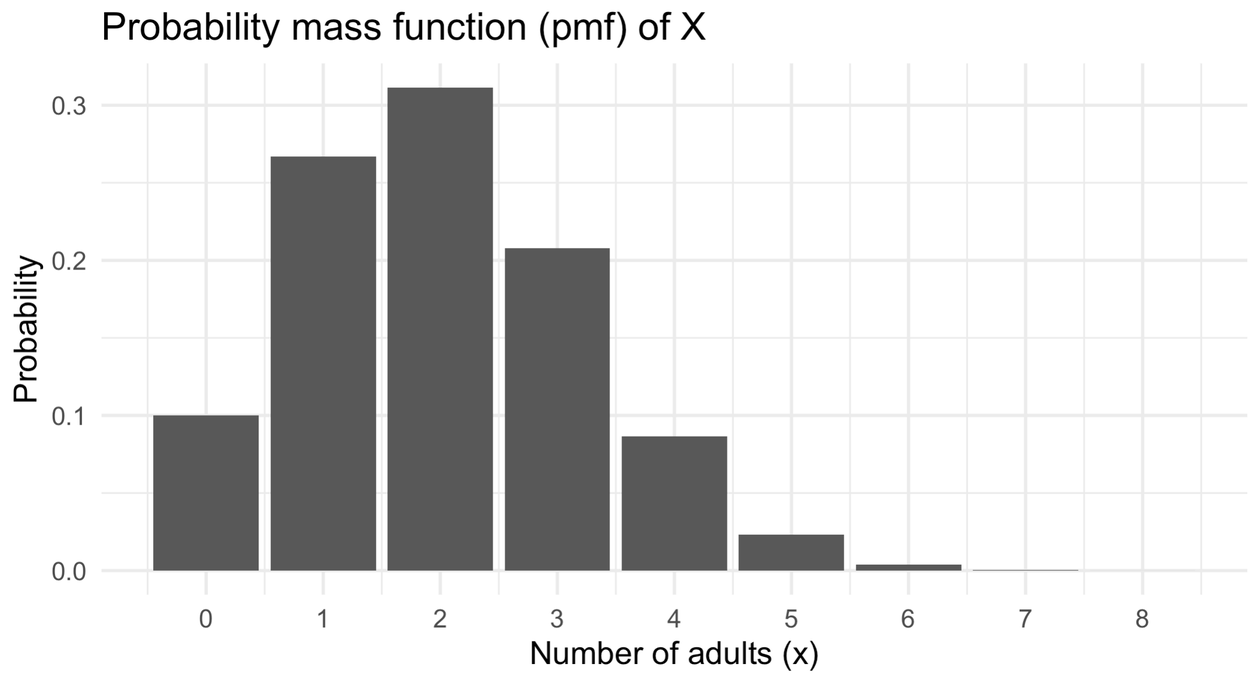
```
# A tibble: 9 × 2
      x      prob
  <int>     <dbl>
1     0 0.100
2     1 0.267
3     2 0.311
4     3 0.208
5     4 0.0865
6     5 0.0231
7     6 0.00385
8     7 0.000366
9     8 0.0000153
```

Falls in Older Adults (5/8)

Example 1: Falls in Older Adults

4. Make a bar plot of the pmf.

- Plot the pmf of X



Falls in Older Adults (6/8)

Example 1: Falls in Older Adults

5. Simulate X for 10000 groups and plot the approximated pmf.

- Act out the story: every person in every group either falls or does not

```
1  reps <- 10000                                                    ①
2
3  people <- expand_grid(sim = 1:reps, person = 1:n) |>             ②
4    mutate(outcome = sample(c("fall", "no fall"),                  ③
5                            size = n(),
6                            replace = TRUE,
7                            prob = c(p, 1 - p)))
8
9  falls_sims <- people |>                                          ④
10   group_by(sim) |>
11   summarise(X = sum(outcome == "fall"))
```

- The first few simulated values of X:

```
1  falls_sims                                                       ①
```

```
# A tibble: 10,000 × 2
     sim     X
   <int> <int>
 1     1     2
 2     2     4
 3     3     0
 4     4     1
 5     5     1
 6     6     1
 7     7     4
 8     8     3
 9     9     4
10    10     1
# ℹ 9,990 more rows
```

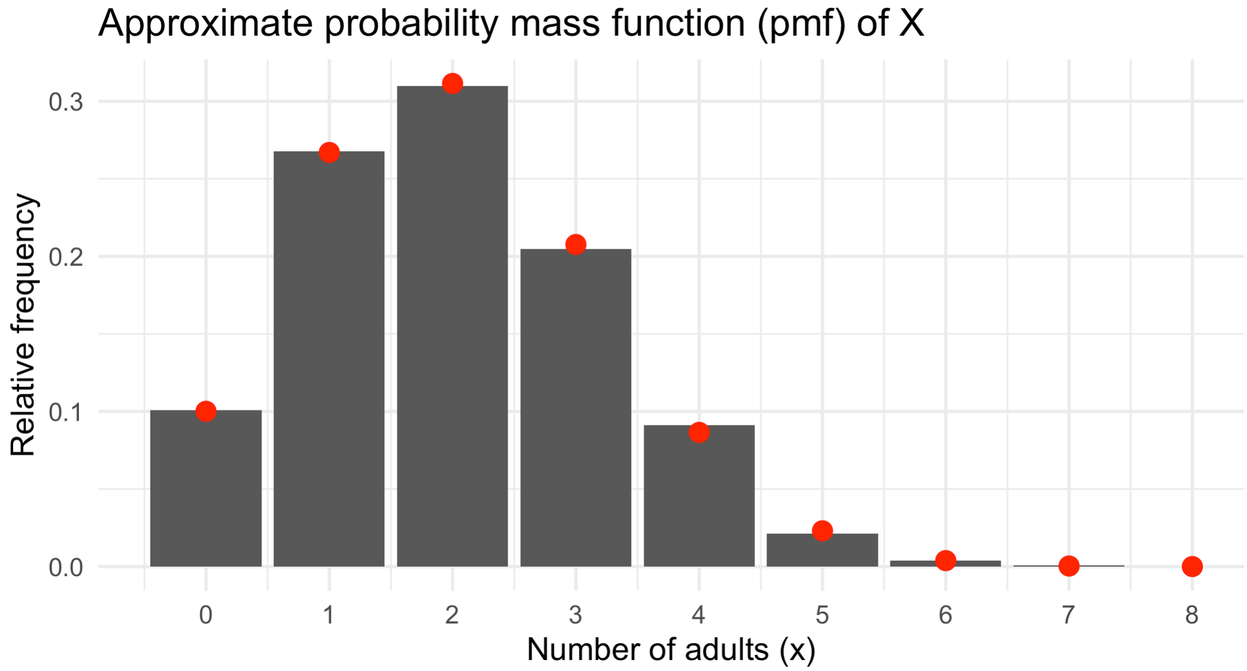
Falls in Older Adults (7/8)

Example 1: Falls in Older Adults

5. Simulate X for 10000 groups and plot the approximated pmf.

- Red dots: the exact pmf from `dbinom()`
- Bars: the approximate pmf from the simulation

▶ Plot the approximate pmf of X



Falls in Older Adults (8/8)

Example 1: Falls in Older Adults

5. Simulate X for 10000 groups and plot the approximated pmf.

- Because X is binomial, `rbinom()` is a shortcut for the whole story above

```
1  falls_both <- falls_sims |>                                          ①
2    mutate(X_shortcut = rbinom(n = reps, size = n, prob = p))          ②
3
4  falls_both |>                                                        ③
5    summarise(box_model = mean(X == 2),                                ④
6              shortcut = mean(X_shortcut == 2),
7              exact = dbinom(2, size = n, prob = p))                   ⑤
```

```
# A tibble: 1 × 3
  box_model shortcut exact
      <dbl>    <dbl> <dbl>
1     0.310    0.314 0.311
```

- All three answers agree (up to simulation error)
- `rbinom()` simulates the same story in one line

Falls in Older Adults: changing the parameters

- Each value of the parameters gives a **different** pmf
- What happens to the pmf of X if the chance of a fall, p, changes?

▶ Plot the pmf for three values of p

